

Institute of Actuaries of India

Subject SP6 – Financial Derivatives Principles

March 2022 Examination

INDICATIVE SOLUTION

Introduction

The indicative solution has been written by the Examiners with the aim of helping candidates. The solutions given are only indicative. It is realized that there could be other points as valid answers and examiner have given credit for any alternative approach or interpretation which they consider to be reasonable

Solution 1:

- i) The value of the trader's position is

$$V_t = \phi_t S_t - C_t$$

V_t is risk-free if and only if the coefficient of randomness dW_t is 0, during a small-time interval dt . Thus, the change in value during dt is

$$dV_t = \phi_t dS_t - dC_t = \phi_t(\mu S_t dt + \sigma S_t dW_t) - (\mu_t^C C_t dt + \sigma_t^C C_t dW_t)$$

Since the trader's position does not change, it is risk free if and only if

$$\phi_t \sigma S_t = \sigma_t^C C_t$$

Thus, the result.

[3]

- ii) No – arbitrage condition implies that the drift in the no – risk position must be the short – term interest rate. Thus,

$$\phi_t \mu S_t - \mu_t^C C_t = r V_t = r(\phi_t S_t - C_t)$$

which, on substituting ϕ_t gives

$$\sigma_t^C C_t \mu / \sigma - \mu_t^C C_t = r(\sigma_t^C C_t / \sigma - C_t).$$

Divide both sides of the above equation by

$$C_t \sigma_t^C.$$

This gives the result. Following on from (b), this implies that there exists a common risk premium for the asset and all derivatives based on the asset. The common risk – premium is given as

$$\lambda = (\mu - r) / \sigma$$

[3]

- iii) Ito's Lemma shows us

$$dC_t = \frac{\partial C}{\partial t} dt + \frac{\partial C}{\partial S} dS + \frac{1}{2} \frac{\partial^2 C}{\partial S^2} (dS)^2$$

The other 2nd order derivatives are 0. Thus, the only non – zero term is $(dS)^2$ is

$$\sigma^2 S_t^2 (dW_t)^2 = \sigma^2 S_t^2 dt.$$

Thus,

$$dC = \left(\frac{\partial C}{\partial t} + \frac{\partial C}{\partial S} \mu S + \frac{1}{2} \frac{\partial^2 C}{\partial S^2} \sigma^2 S^2 \right) dt + \sigma S \frac{\partial C}{\partial S} dW_t$$

The relationship can then be proven by identifying $C_t \mu_t^C$ with the coefficient of dt and $C_t \sigma_t^C$ with the coefficient of dW_t .

[4]

- iv) Following on from (b), we get

$$\mu_t^C = r + \sigma_t^C (\mu - r) / \sigma = r + (S/C) \frac{\partial C}{\partial S} (\mu - r)$$

Substitute this in (c) to get

$$\mu^C = C^{-1} \left(\frac{\partial C}{\partial t} + \mu S \frac{\partial C}{\partial S} + \frac{1}{2} \sigma^2 S^2 \frac{\partial^2 C}{\partial S^2} \right) = r + (S/C) \frac{\partial C}{\partial S} (\mu - r)$$

Hence, the result. This means that derivatives can be priced without knowing the rates of return on the asset.

[4]

[14 Marks]

Solution 2:

- i) **Call futures options**- The holder receives a long position in the underlying futures contract plus an amount of cash equal to the current futures price minus the strike price. On the other hand, **Put futures options** - The holder gets a short position in the underlying futures contract plus an amount of cash equal to the difference between the strike price and the current futures price.

[2]

- ii) Suppose there is constant risk-free interest rate r and the futures price $F(t)$ of a particular underlying is log-normal with constant volatility σ . Then Black's formula states the price for a European call option of maturity T on a futures contract with strike price K and delivery date T' (with $T' \geq T$) is

$$c = e^{-rT} [FN(d_1) - KN(d_2)]$$

The corresponding put price is

$$p = e^{-rT} [KN(-d_2) - FN(-d_1)]$$

[1]

- iii) Using the Black's formula to calculate the value of option

d1 =	-0.578824
d2 =	-0.731576
N(d1) =	0.2813607
N(d2) =	0.2322192
Put =	6906

[3]

[6 Marks]

Solution 3:

- i) Apply the Black Scholes formula given by $C = SN(d_1) - X e^{-rT} N(d_2)$. Substituting the numbers, this works out to 2.374

$$\text{Delta} = N(d_1) \text{ and } \text{Gamma} = \frac{N'(d_1)}{S\sigma\sqrt{T}}, \text{ where } N'(d_1) = \frac{1}{\sqrt{2\pi}} e^{-\frac{d_1^2}{2}}.$$

Substituting the values, we get Delta = 0.562 and Gamma = 0.0747.

[4]

- ii) (a) Delta hedging requires the trader to constantly watch and adjust the positions involved. This can lead to significant transactional costs and affect the portfolio returns. If the

rebalancing is done only after a particular quantum of price movement to reduce the trading costs the hedging will not be perfect. The trade off between frequency of doing the rebalancing and the transaction cost is one major drawback of delta hedging.

(b) Delta hedge is effective for small price ranges. Situations may so arise where there is a massive change in prices at the opening itself as markets react to overnight developments which are either market specific or stock specific. Under such circumstance even a perfect overnight delta hedge will become ineffective in achieving the desired purpose.

[3]

- iii) To ensure delta neutrality, he needs to buy shares and thus borrow money. Let us say he buys "n" shares and borrows an amount B. The following two equations need to be satisfied

Portfolio has to be self-financing: $1000(2.374) - n(50) - B = 0$

Portfolio has to be delta neutral: $1000(0.562) - n(1) = 0$

This implies the number of shares to be bought is 562 and the amount of money to be borrowed is this \$25726

[4]

- iv) If the stock price moves to \$52, the delta will have to be recalculated. The new delta for the updated price is 0.7017. This implies that the total number of shares required to be delta neutral on this day is 701.7 or approximately 702. Thus, the writer needs to buy an addition 139.7 or approximately 140 shares to remain delta neutral.

[4]

- v) To ensure delta neutrality, he needs to buy shares, options and thus borrow money. Let us say he buys "n₁" shares, "n₂" options and borrows an amount B. The following two equations need to be satisfied

Portfolio has to be self-financing: $1000(2.374) - n_1(50) - n_2(1.1466) - B = 0$

Portfolio has to be delta neutral: $1000(0.562) - n_1(1) - n_2(0.2965) = 0$

Portfolio has to be gamma neutral: $1000(0.0747) - n_2(0.0529) = 0$

- a) This implies that the number of new options (option 2) that he has to buy is 1412.10 or approximately 1412. This implies the number of shares to be bought is 143.31 or approximately 143 and the amount of money to be borrowed is this \$6410.75.

[3]

- b) The reason the amount of money required is less is because of the presence of a large number of options where the current outflow is the premium only.

[2]

[20 Marks]

Solution 4:

- i) Using the expected return and volatility for LUH, the market price of risk for Nifty 50 is:

$$\frac{\mu_1 - r}{\sigma_1} = \frac{0.20 - 0.06}{0.25} = 0.56$$

[3]

- ii) When there is just one source of randomness, derivatives on the same underlying asset would have the same market price of risk; i.e.

$$\frac{\mu_1 - r}{\sigma_1} = \frac{\mu_2 - r}{\sigma_2} = 0.56$$

$$\Rightarrow \mu_2 = (0.56 * 0.35) + 0.06 = 0.256$$

Thus, the expected return from G&P is 25.6%.

[2]

[5 Marks]

Solution 5:

- i) Probability of ~~D~~Zee Telefilms defaulting is 3.6%
 Probability of ~~Vertex~~Axis Bank defaulting is 0.5%
 Given that each of the events are independent of each other, and that ~~CFDHHDFC~~ Bank will suffer a loss only if both these events happen together.

The probability of ~~CFDHHDFC~~ Bank incurring a loss is the product of the probability of default of ~~Zee~~Dee Telefilms and the probability of default of ~~Axis~~Vertex Bank = 0.5% * 3.6% = 0.018%

[3]

- ii) The expected loss, given a \$100 million exposure with a probability of 0.018% is
 $\$100 \text{ million} * 0.018\% = \$18,000$

[2]

[5 Marks]

Solution 6:

- i) Swaption is option to enter swap at a future date with strike swap rate. The company should enter into a receiver swaption which enter in to receiver swap to pay floating and receive Fixed amount.

[3]

- ii) As a first step estimate the forward swap rate

$$\text{Bond (11)} - \text{bond (15)} / (\text{Sum bond value 11 -15})$$

$$= 5.41\%$$

Using Blacks formula estimate the valuation of the swaption

Number of payments/year	1	npay
Coupon Payment	600000 0	P
10 Year Forward Swap Rate	5.4%	Forward swap rate
Strike Rate	6.0%	Strike rate
Volatility of Forward Rate	25.0%	volatility
Risk Free Rate	5.3%	
Results		
Receiver	311,626. 741	=Principle * Annuity factor * EXP(-rfr * T) * (Fixed * nndtwo - ForwardSwap * nndone)
Intermediate Calculations		
Swap Tenor (years)	5	
Swap Rate Period (years)	10	
dt	0.79056 942	Volatility * Sqrt(10)
d1	0.26513 383	(LN(ForwardSwap / Fixed) + (0.5 * V ^ 2) * T) / dt
d2	- 0.52543 558	-d _{one} -d _t
nd1	0.60454 682	-NORMSDIST(d _{one})
nd2	0.29964 021	-NORMSDIST(d _{two})
nnd1	0.39545 318	-NORMSDIST(-d _{one})
nnd2	0.70035 979	-NORMSDIST(-d _{two})
A	4.28045 164	-((1 - (1 / ((1 + (ForwardSwap / npay)) ^ (T_1 * npay)))) / ForwardSwap)

[5]

[8 Marks]

Solution 7:

i) Bond A default calculation

correlation	25%		Price allowing for default	No default price	Default allowance		
recovery	40%		94.1043084	100	5.89569161		
Bond A						end of year 2	
	Principle	Default during year	Principle end of year	coupon	recovery	cashflow	Discount
0	100	5	95	4.75	2	6.75	6.42857143
1	95	4.75	90.25	4.5125	1.9	96.6625	87.675737

Bond B

			Price allowing for default	No default price	Default allowance		
Bond B			85.8704165	100	14.1295835		
	Principle	Default during year	Principle end of year	coupon	recovery	cashflow	Discount
0	100	10	90	5.4	2.5	7.9	7.45283019
1	90	9	81	4.86	2.25	88.11	78.4175863

BOND A	5.90
BOND B	14.13
TOTAL	20.03

[5]

ii)

We would not expect both the bonds to default at the same time hence using correlation based approach may be useful.

Key downsides.

- 1) If both the bonds are issued by the same provided, then defaults can happen at the same time.
- 2) Bond from similar industry or sector behave in similar fashion
- 3) During the extreme market scenarios, the correction can be close to 1.

[4]

iii)

Using 2 variable correction matrix the capital requirement would be

$$\text{div Capital} = \sqrt{\text{cap } 1^2 + \text{cap } 2^2 + 2 \cdot \rho \cdot \text{cap } 1 \cdot \text{cap } 2}$$

Total allowing for diversification	16.61
Total saving compared to previous scenario	3.41

[3]

[12 Marks]**Solution 8:**

- i) In unexpected high inflation scenario bond issuer who is receiving the money now and paying back in future via interest and principal payments –would benefit as due to unexpected inflation future value of the money paid would be lower in real term. [2]
- ii) Inflation can be hedged using index linked bonds where the payment on the bonds is linked to price index

The inflation bonds may not be easily available for all duration

There can be credit risk associated with the inflation bond if not issued by the Government

There are option to enter in to LPI/RPI swaps which would provide inflation hedge without exchange on nominal amount.

The loss can be floored by Floor and CAP

Valuation of these instruments are not easy, the market for these instruments is also limited [3]

iii)

Valuation						
	Un adjusted coupon	Final payment	Adjusted coupon	Adj. final payment	Total payment	Accumulation value
1	3	0	3.25	0	3.25	3.45
2	3	0	3.38	0	3.38	3.61
3	3	100	3.75	125.00	128.75	128.75
						135.81
	Return over 3 years	36%				

Use the interest rate table to get accumulation interest rate with assumption that money can be reinvested at the n year bond rate.

Real interest rate table is not used.

[5]

[10 Marks]

Solution 9:

i) Calculation of the bond is shown in the table below.

		Scenario A	104.33	Scenario B	100.00	Scenario C	95.90
	interest rate	5%	interest rate	6%	interest rate	7%	
S.No	Cashflow	1.00		2.00		3.00	
1	6	5.71		5.66		5.61	
2	6	5.44		5.34		5.24	
3	6	5.18		5.04		4.90	
4	6	4.94		4.75		4.58	
5	6	4.70		4.48		4.28	
5	100	78.35		74.73		71.30	

Scenario A	Premium	4.329%
Scenario B	At par	0%
Scenario C	Discount	-4.10%

[4]

ii)

	Running yield = Annual coupon/Price	GRY (Same as the fixed interest rate)
Scenario A	5.7510%	5%
Scenario B	6.0000%	6%
Scenario C	6.2565%	7%

Running yield is lower due to higher initial value of the bond compared to cashflow.

[4]

iii)

	Scenario C	7%	
S.No		Price	96.61
1	6.00	5.61	
2	6.00	5.24	
3	6.00	4.90	
4	6.00	4.58	
4	100.00	76.29	

Explanation of movement

t=0 value	95.90
T= 1 value	96.61
Return in year (GRY x bond price t=0)	6.71
Coupon	6
Excess return	0.71
Change in valuation	0.71
Hence excess return is equal to change in valuation	

[4]

[12 Marks]

Solution 10:

- i) The announcement will have significant ramifications on the capital allocation policy of the listed companies. Dividends getting taxed at the hands of the investor and majority of the companies being promoter driven would mean that after three months the dividends that the shareholders receive in hand after taxes would be much lower. Say the marginal rate of tax at highest slab is 30% then shareholders will now only get 70% of the declared dividend in hand. This can lead to many companies immediately distributing large amounts of surplus reserves as onetime special dividends, to make it a tax efficient distribution for the promoters and other shareholders. XYZ has been a mature market and companies in mature markets usually follow a regular distribution policy – say a fixed percentage of PAT policy, low but steady increasing stream policy, one final dividend or a few interim dividends and so on. So, market participants are generally aware of the expected dividends and adjust the derivative prices accordingly. The impact of ex-dividend date falling in before the expiry of a futures or option contract is built into the price. The current announcement will lead to a huge disruption as large special dividends can have a significant impact on the options and futures payoff if unadjusted. This can lead to chaos in the derivative market. There may be a lack of liquidity on fresh positions as traders stay away. The traders who are already having open positions on long dated derivatives may not have anticipated this announcement and could lead to large losses/gains depending on which leg of the trade they are exposed to. The confidence in the system may be affected.

[4]

- ii) Ideally the traders should not be making large gains or losses due to such policy changes. Hence the parameters of the derivative should be adjusted for such actions. In case of options the policy should be amended that the full value of dividend is deducted from the cum dividend strike prices as on the ex-dividend date. For futures the adjustment has to be made on the MTM calculations. The future price on the ex-dividend date will open lower tracking the spot price. However, the last cum-dividend close price less the dividend can be considered for the ex-date MTM calculations. Lot-sizes need not be changed. After the three months the companies will adjust to new regime and markets will start making appropriate expectations.

But adjusting for every regular dividend like this will be cumbersome. So, the exchanges can probably set a threshold say a dividend that leads to a dividend yield of more than 4% or any other threshold need to be considered for adjustments. The tax regime may also lead to companies resorting to other capital allocation policies. Say, they may start preferring a buyback rather than a dividend. The quantum of buyback, price at which buyback will be done, mode in which it will be done say a tender offer or market operation and so on can also impact the spot prices and hence the derivatives. But over time, once companies declare their allocation strategy to investors the markets will soon start making informed expectations. No further adjustment may hence be needed.→

[4]

[8 Marks]
